

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-3

DURATION: 30 MINUTES

SUMMER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. In Figure 1, the area of the shaded regions A and B , portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively, are equal. The aforementioned functions are,

5

(CO2)

(PO1)

$$f_a(x) = -x^2 + k$$

$$f_b(x) = x^3 + x^2$$

Using the Composite Trapezoidal Rule, with $n = 8$ segments, approximate the value of k .

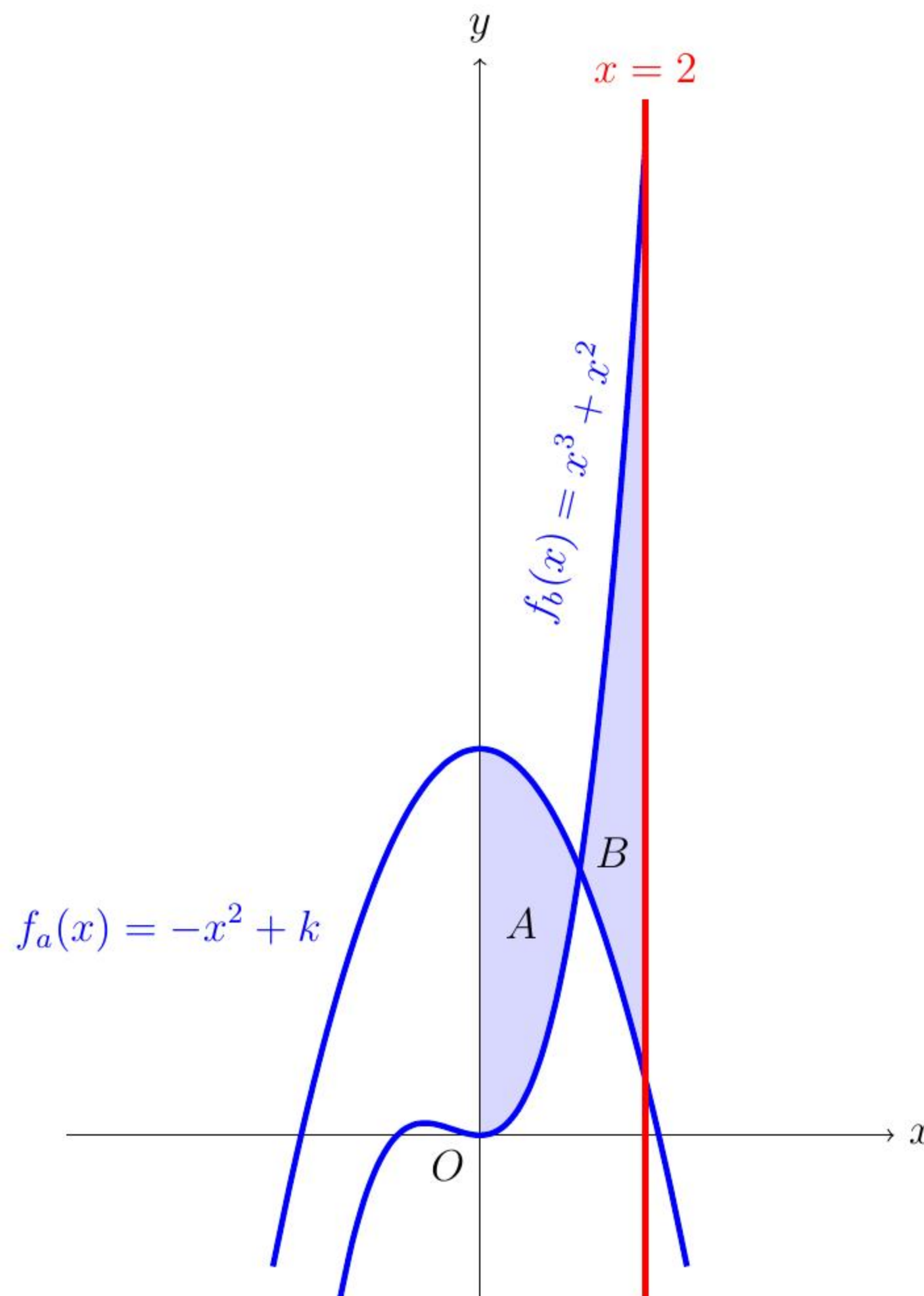


Figure 1: The shaded regions A and B .

2. Suppose, you are given a set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and you want to regress the data to an m th order Polynomial model, where $m < n$. How would you choose a suitable value of m ? Explain with proper illustration(s).

4

(CO3)

(PO1)

3. Suppose, you are monitoring the progress of a homogeneous chemical reaction in order to calculate the rate constant k and the order of the reaction n . The rate law expression for the reaction can be represented with the Power model 5
(CO2)
(PO1)

$$-r = kC^n$$

where r is the reaction rate and C is the concentration of the reactant. The values of reaction rate $-r$ and the respective concentration C of the reactant that you have calculated over the course of time are shown in the table below –

Concentration, C ($gmolL^{-1}$)	Reaction Rate, $-r$ ($gmolL^{-1}s^{-1}$)
4	0.398
2.25	0.298
1.45	0.238
1	0.198
0.65	0.158
0.25	0.098
0.006	0.048

Table 1: Chemical kinetics of the observed homogeneous reaction.

Determine the value of the rate constant k and the order of the reaction n using the data provided in Table-1. You are allowed to apply data transformation.

4. What is Regression Analysis? Provide a precise formulation of the problem statement of a regression model. 1
(CO2)
(PO1)